Dual Spacetime 4-Valued Logic (D4L): Amplitudes, Four States, Geometric Interference, and Dual Hilbert Space

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Abstract

Classical two-valued logic is the logic of a decided world: a proposition is true or false, and a contradiction entails everything. Dual Spacetime Theory is not a theory of decided worlds. Every massive configuration is a pair of frames, usual and dual, and the invariant of that pair is a single signed height $J_{\text{norm}} \in [-1, 1]$. Dual Spacetime 4-Valued Logic (D4L) is the logic of that height.

A proposition is an admissible torsion configuration. Measurement reads J_{norm}; the four names $|T\rangle, |U\rangle, |F\rangle, |B\rangle$ are the sign and the saturation of the same number, not an external algebra of labels. Negation is the usual–dual swap of the geometry. Conjunction and disjunction are min and max on the height. A configuration and its swap cannot both sit at the positive wall, so contradiction does not explode.

The dual sector of the same geometry carries a Hilbert space $\mathbb{C}^2$. Amplitude operations and Hilbert operations belong to one logic, related by an internal dictionary, but they are not identified: the Killing form is indefinite, self-overlap is not a Born probability, and the four labels are not a PVM of $\mathbb{C}^2$. Two designated predicates—synchrony and non-refutation—turn the height into a consequence relation. Two-valued restrictions of the interval fail to host the situations that relation is built to record. A classical tautology is precisely a formula that returns the positive wall under every assignment of atoms to the two walls. The De Morgan skeleton of a classical proof is an identity of every real height, so the two-valued case split is a specialisation; excluded middle and the syllogism are the saturation of the height. Material implication detaches the positive wall on every real height. It lies below that wall exactly when the negated antecedent and the consequent do, and balance implies every nonpositive height. In the spinor lattice, inclusion is the identity that $A^{\perp} \vee (A \wedge B)$ is the whole space. That implication agrees with the classical expansion $A^{\perp} \vee B$, and the expansion detaches, exactly when the antecedent splits along the consequent. The implication carried by the lattice detaches for every pair. The statements of this paper are machine-checked.

1 Introduction

Two-valued logic is the natural language of a fact that has already been decided. A proposition is true or it is false. If it is both, then every other proposition follows: that is the explosion principle, and it is why classical consequence cannot tolerate a live contradiction.

Dual Spacetime Theory does not describe decided facts. It describes a pair of Minkowski frames carried by every massive particle—the usual frame, in which we measure, and a dual frame, time-reversed relative to the first [1, 2]. On each of the three spatial axes the pair is encoded by two rapidities (α_a, β_a): a boost in the usual sector and a rotation in the dual sector. The already-proved invariant of that pair is a signed height. When the two frames are perfectly balanced the height vanishes. When the usual sector dominates, the height is positive. When the dual sector dominates, it is negative. At the walls the configuration is saturated: every axis is a pure boost, or every axis is a pure rotation.

Those four situations are not extra labels glued onto an algebra. They are the sign and the saturation of a quantity the geometry already supplies. Dual Spacetime 4-Valued Logic is the decision to take that quantity as the observable of a proposition, and to let the four names be what the observable does.

The resulting logic has two layers, and they must not be confused.

The *amplitude layer* takes an admissible configuration as a proposition, the normalised height J_{norm} as its observable, and $\min / \max / -$ as scalar connectives. Overlap of two configurations is the parameter Killing form already present in the PGA core. Interference is the geometric commutator of their torsion bivectors. Nothing is added to the algebra; the logic reads what the algebra already computes.

The *dual Hilbert layer* is the kinematics of the dual sector: a Euclidean pairing on $\mathbb{R}^3$, the quaternion table of the cyclic generators, and their representation on $\mathbb{C}^2$. The lattice of closed subspaces is the Birkhoff–von Neumann logic of a two-dimensional spinor [3]. It is a genuine quantum logic. It is not the amplitude layer in another notation.

An internal dictionary relates the layers. Their operations are not identified. The Killing form is not the spinor inner product; min / max is not the meet and join of $L(\mathbb{C}^2)$; the four labels are not a projection-valued measure. There is no $\hbar$, and there is no derivation of the Born rule from J.

The paper develops this picture in that order. Section 2 fixes the geometric carrier and the bound that makes four states possible. Section 3 names the states and splits the balanced label by a second observable, mass, without adding a fifth name. Sections 4 and 5 give the scalar connectives, the two orders, and the geometric operations. Section 6 extracts the dual Hilbert layer and states the dictionary that refuses to identify it with the first. Section 7 says which heights hold. Section 8 separates a classical proof into a height identity, a saturation on the walls, and the three cuts of material implication. Section 9 shows why two-valued restrictions of the interval cannot host the situations D4L is built to record. Section 10 names the splitting on which a classical expansion detaches in the spinor lattice. Section 11 reads DST proof artefacts and Diophantine regimes in the same four names. Section 13 states the reach of the theory, and its limits.

2 Torsion as the carrier of a proposition

A proposition in D4L is not a sentence. It is a configuration of the geometry: six rapidities, equivalently a torsion bivector and the rotor it generates. The observable is a single scalar already proved in the companion development [2].

Write $\Omega = R^{\dagger}_{\text{usual}} R_{\text{dual}}$ for a rotor pair, and let Ω_{biv} be a principal logarithm. Throughout this paper the parameter form of the torsion scalar is

$$J = \frac{1}{2}\sum_{a=1}^{3}(\alpha_a^2 - \beta_a^2). \tag{1}$$

The combination $\frac{1}{16}B(\Omega_{\text{biv}}, \Omega_{\text{biv}})$ that appears in older drafts differs from (1) by a constant factor under the standard generator expansion. Boundedness statements here always refer to J and to its normalisation J_{norm}.

Not every six-tuple of rapidities is admitted. Usual and dual rapidities on one axis are not allowed to add past a right angle: that is the anti-synchronisation bound of the dual map. Together with non-negativity it cuts out a compact cone.

Definition 2.1 (Admissible configuration)**.** A configuration is *admissible* when, on every axis,

$$\alpha_a \geq 0, \qquad \beta_a \geq 0, \qquad \alpha_a + \beta_a \leq \frac{\pi}{2}.$$

On that cone the signed height cannot run off to infinity. The quadratic form J is bounded by the volume of the cone, and a constant rescaling puts the bound at 1.

Theorem 2.2 (Torsion bound)**.** *On the admissible cone of Definition 2.1,*

$$|J| \leq \frac{3\pi^2}{8}, \qquad |J_{\text{norm}}| \leq 1,$$

where

$$J_{\text{norm}} := \frac{8}{3\pi^2}\, J. \tag{2}$$

Equality $|J_{\text{norm}}| = 1$ holds if and only if every axis is a pure boost, or every axis is a pure rotation. The image of J_{norm} on the admissible cone is the whole interval $[-1, 1]$. The bound is false if one keeps only the principal-branch condition $|\alpha_a + \beta_a| \leq \pi/2$ and allows negative rapidities.

The four names of D4L are therefore not a truncation. Every value in $[-1, 1]$ is attained, and both walls are attained, but only by configurations in which every axis votes the same way. A mixed extremum—one axis a pure boost, another a pure rotation—never reaches $|J_{\text{norm}}| = 1$.

On a compactification N the discrete bound is sharper [2]:

$$|J_{\text{norm}}| \le \Big(\frac{4\lfloor N/4\rfloor}{N}\Big)^2.$$

When $4 \mid N$ the right-hand side is 1 and both signs are attained. When $4 \nmid N$ every admissible lattice point satisfies the strict inequality $|J_{\text{norm}}| < 1$: the positive wall is simply not there.

An *amplitude* is an admissible continuous configuration $p = (\alpha_a, \beta_a)$. Its torsion bivector and rotor are

$$\Omega(p) = \sum_{a=1}^{3}\Big(\frac{\alpha_a}{2}\,B_a^+ + \frac{\beta_a}{2}\,B_a^-\Big), \qquad R(p) = \exp\Omega(p).$$

Measurement and collapse are

$$\langle p\rangle := J_{\text{norm}}(p), \qquad \text{collapse}(p) \in \{|T\rangle, |U\rangle, |F\rangle, |B\rangle\}.$$

The adjoint is the usual–dual swap $p^\dagger$, exchanging (α_a) with (β_a). The swap is an involution and flips the observable: $(p^\dagger)^\dagger = p$ and $\langle p^\dagger\rangle = -\langle p\rangle$. Every label is realised by some amplitude.

3 Four states from a signed height

Once J_{norm} is known to take every value in $[-1, 1]$, the four names are a partition of that interval by sign and saturation.

Definition 3.1 (D4L states)**.** On $J_{\text{norm}} \in [-1, 1]$,

$\lvert T\rangle$: $J_{\text{norm}} = 0$	perfect usual–dual balance,
$\lvert U\rangle$: $0 < J_{\text{norm}} < 1$	boost-dominant, not saturated,
$\lvert F\rangle$: $J_{\text{norm}} = +1$	every axis a pure boost,
$\lvert B\rangle$: $-1 \le J_{\text{norm}} < 0$	rotation-dominant, including the deepest point $J_{\text{norm}} = -1$.

The positive extreme is a singleton named False. The negative extreme is kept inside Both. That is a logical convention, not a gap in the cone: both walls exist and are unique. On an admissible configuration, collapse to $|F\rangle$ if and only if the configuration is purely hyperbolic, and $J_{\text{norm}} = -1$ if and only if it is purely elliptic.

Interior $|B\rangle$ is a range, not a single fixed point. The deepest point of that range is the wall $J_{\text{norm}} = -1$. The four names are therefore not a function of each other under adjoint: deepest $|B\rangle$ maps to $|F\rangle$, while interior $|B\rangle$ maps to $|U\rangle$. Negation, as a map on labels, does not exist.

The balanced label $|T\rangle$ is likewise not a single geometric point. The unsigned torsional mass

$$M(p) = \frac{1}{2}\sum_{a=1}^{3}(\alpha_a^2 + \beta_a^2), \qquad M_{\text{norm}} = \frac{8}{3\pi^2}\,M$$

is a second observable, not a fifth name. On the admissible cone $0 \le M_{\text{norm}} \le 1$. Usual–dual swap preserves M; uniform scaling multiplies M by k^2. Height cannot exceed mass: $|J| \le M$, equivalently $|J_{\text{norm}}| \le M_{\text{norm}}$. Equality holds if and only if one sector vanishes entirely—every dual rapidity is zero, or every usual rapidity is zero. Saturation of height relative to mass is purity of sector.

Vanishing height is still not a single geometric condition. The origin of the cone is vacuum. Axiswise equality $\alpha_a = \beta_a$ on every axis is balanced massive. There is a third seat: a boost on one axis can cancel a rotation on another, so that $J_{\text{norm}} = 0$ and $M > 0$ without axiswise equality. The four-state classifier sees only the height, and cannot tell these three apart.

Vacuum and balanced massive are predicates on amplitudes:

$$\begin{aligned} \text{IsVacuum}(p) &\iff \text{collapse}(p) = |T\rangle \wedge M(p) = 0, \\ \text{IsBalancedMassive}(p) &\iff \text{collapse}(p) = |T\rangle \wedge 0 < M(p). \end{aligned}$$

Vacuum is equivalent to all six rapidities vanishing, and implies $J_{\text{norm}} = 0$. The converse is false. Beal's balanced seeds sit in the axiswise seat; the cross-axis seat is a different geometric reason for the same label.

4 Connectives that do not explode

The scalar operations on the signed height are the ones the real line already possesses:

$$\neg j = -j, \qquad j \wedge k = \min(j,k), \qquad j \vee k = \max(j,k).$$

Negation is not a new connective. It is the usual–dual swap $(\alpha_a, \beta_a) \mapsto (\beta_a, \alpha_a)$, which sends $J \mapsto -J$ and therefore $J_{\text{norm}} \mapsto -J_{\text{norm}}$. De Morgan duality and distributivity hold as they do for min / max on $\mathbb{R}$.

The logical content is in what conjunction of a height with its negation cannot do.

Theorem 4.1 (Non-explosion)**.** *For every $j \in [-1,1]$,*

$$j \wedge \neg j = -|j| \leq 0.$$

Hence $p \wedge \neg p$ is $|T\rangle$ or $|B\rangle$, and never saturates $|F\rangle$.

The dual identity is $j \vee \neg j = |j|$. Excluded middle is synchronous only at balance, and on the interval it is refuted exactly at the two walls.

A configuration and its usual–dual swap cannot both sit at the positive wall. Contradiction is therefore a rotation-dominant, or balanced, situation. It is never the unique all-boost world. That is why D4L can host a live contradiction without forcing every unrelated atom onto $|F\rangle$.

The interval $[-1,1]$ carries two preorders, and they must not be collapsed into one. The *height* order is the ordinary order of the real line: $j \preceq_\tau k$ if and only if $j \leq k$. It compares boost-dominance with rotation-dominance. The *information* order compares saturation: $j \preceq_\iota k$ if and only if $|j| \leq |k|$. In the information order, $j = 0$ is the unique bottom, and the two tops are exactly the walls $j = \pm 1$. Equivalently, the information bottom is the label $|T\rangle$, and the two tops are $|F\rangle$ and deepest $|B\rangle$. Interior $|B\rangle$ is strictly below deepest $|B\rangle$ in the information order.

The four labels therefore do not match Belnap's $\{\bot, t, f, \top\}$. No bilattice isomorphism is claimed. Height and saturation are two readings of one number, not two independent lattices.

5 Overlap and interference

Scalar connectives act after measurement. Before measurement, two amplitudes already interact in the algebra. The commutative interaction is overlap; the non-commutative interaction is interference.

$$\langle p \mid q \rangle := 8 \sum_{a=1}^{3} \big(\alpha_a \alpha'_a - \beta_a \beta'_a\big), \qquad [\Omega(p), \Omega(q)] := \Omega(p)\Omega(q) - \Omega(q)\Omega(p).$$

Rotor composition $R(p)R(q)$ is a product in the PGA. It is not pulled back through the Baker–Campbell–Hausdorff formula to a third torsion configuration.

The overlap is symmetric, and self-overlap is a constant multiple of the observable: $\langle p \mid p \rangle = 16\, J(p)$, equivalently a constant multiple of $J_{\text{norm}}(p)$. Distinct-axis torsion bivectors need not commute. If p is a pure boost on axis 0 and q a pure rotation on axis 1, then $[\Omega(p), \Omega(q)] \neq 0$. Same-axis hyperbolic and cyclic generators do commute, so the corresponding one-axis interference vanishes. The converse is false: vanishing interference does not force a common single axis. A two-axis boost, for instance, has vanishing self-commutator and is not single-axis.

Because self-overlap is a multiple of J, it inherits the signature of J. Pure-boost extremals give $\langle p \mid p \rangle > 0$; pure-rotation extremals give $\langle q \mid q \rangle < 0$. The pairing is therefore indefinite. A signed quantity cannot be a Born probability, and the pairing is not the spinor inner product of Section 6.3. Sandwich evaluation of rotors is independent of the gravity chart; preservation of the degenerate quadratic form is not claimed. Geometric composition is associative as a PGA product and is not a lattice operation.

6 The dual sector as a Hilbert space

The dual sector already contains a Euclidean three-space, a quaternion table, and a spinor representation on $\mathbb{C}^2$. This section reads that kinematics as the second layer of D4L, and states the dictionary that relates it to the first. Spatial axes are indexed by $a = 0, 1, 2$.

6.1 Two definite sectors

A configuration is dual-sector when $\alpha = 0$, usual-sector when $\beta = 0$. Write p_β and p_α for the corresponding embeddings of Euclidean $\mathbb{R}^3$. On the dual sector the Killing overlap restricts to

$$\langle p_\beta \mid p_\gamma \rangle = -8\langle \beta, \gamma \rangle_{\mathbb{R}^3},$$

hence is negative definite. On the usual sector it restricts to $+8\langle \alpha, \gamma \rangle_{\mathbb{R}^3}$, hence is positive definite. The inner-product space is the linear span of each sector, not the admissible cone $\alpha_a, \beta_a \geq 0$, $\alpha_a + \beta_a \leq \pi/2$. Superposition in that span is the internal sum, which has no ceiling. The expressed mismatch is the point of the cone closest to the sum.

6.2 Quaternion table

Write Γ_a for the cyclic generators of the dual sector. They satisfy $\Gamma_a^2 = -1$ and the off-diagonal products of the quaternion algebra:

$$\begin{aligned} \Gamma_0\Gamma_1 &= \Gamma_2, & \Gamma_1\Gamma_0 &= -\Gamma_2, \\ \Gamma_1\Gamma_2 &= \Gamma_0, & \Gamma_2\Gamma_1 &= -\Gamma_0, \\ \Gamma_2\Gamma_0 &= \Gamma_1, & \Gamma_0\Gamma_2 &= -\Gamma_1. \end{aligned}$$

The identities follow from the vector anticommutators in $G(3,1,1)$. For a pure dual configuration

$$\Omega(p_\beta) = \sum_a (\beta_a/2)\,\Gamma_a,$$

and on axis 0 this collapses to $(\theta/2)\,\Gamma_0$.

6.3 Spinor Hilbert space

The dual spinor space is $\mathbb{C}^2$ with the standard Hermitian product. Pauli matrices $\sigma_x, \sigma_y, \sigma_z$ are the usual ones. The representation $\rho(\Gamma_a) := -i\sigma_a$ is chosen so that $IJ = K$ agrees with the cyclic products in $G(3,1,1)$ and so that the dual rotor is sent to the standard SU(2) factor $\exp(-i\beta \cdot \sigma/2)$. Each $\rho(\Gamma_a)$ squares to -1, the cyclic identity $\rho(\Gamma_0)\rho(\Gamma_1) = \rho(\Gamma_2)$ holds, and each generator is anti-Hermitian.

The dual rotor matrix is therefore

$$U(\beta) = \exp\Big(\sum_a (\beta_a/2)(-i\sigma_a)\Big) = \exp(-i\beta \cdot \sigma/2).$$

On a single axis this is the Rodrigues formula $\cos(\theta/2)\,I + \sin(\theta/2)\,(-i\sigma_a)$. For a general dual rapidity the same trigonometric form holds along the unit cyclic combination determined by β.

Theorem 6.1 (Dual rotor in SU(2))**.** *For every dual rapidity $\beta \in \mathbb{R}^3$, the matrix $U(\beta)$ is unitary of determinant 1. Thus $U(\beta) \in \mathrm{SU}(2)$. On a pure dual configuration the Banach exponential in $G(3,1,1)$ collapses to the same Rodrigues formula, so the dual-sector exponential and the matrix rotor agree on every pure dual rapidity.*

The usual sector is the twin, with opposite signature. Distinct hyperbolic generators anticommute and each squares to $+1$, so the combination $\Gamma^+(\alpha) = \sum_a \alpha_a B_a^+$ satisfies $\Gamma^+(\alpha)^2 = \|\alpha\|^2$. The Banach exponential on a pure usual rapidity is therefore the hyperbolic Rodrigues formula

$$\exp\big(\Omega(p_\alpha)\big) = \cosh(\|\alpha\|/2)\,I + \sinh(\|\alpha\|/2)\,U^+(\alpha),$$

and on axis 0 this is $\cosh(\theta/2)\,I + \sinh(\theta/2)\,B_0^+$. Representing the same generators by Pauli matrices rather than $-i\sigma_a$ sends the usual rotor to

$$V(\alpha) = \exp(\alpha \cdot \sigma/2).$$

For every usual rapidity, $V(\alpha)$ is Hermitian of determinant 1, and the matrix exponential agrees with the Banach exponential on every pure usual configuration. A nonzero boost is not unitary: the usual rotor is not an element of SU(2). The set of Hermitian determinant-1 matrices is not claimed to be a group under multiplication.

This is the kinematics of the positive wall. Height equals mass if and only if one sector vanishes; the usual exponential saturates $|F\rangle$ as the dual exponential saturates deepest $|B\rangle$.

A general identification of the Banach exponential in $G(3,1,1)$ with this matrix exponential for *mixed* usual/dual or translational generators is not claimed; that would require control of the Baker–Campbell–Hausdorff formula beyond a single pure sector. The product of a usual boost with a dual rotation is not identified with an $\mathrm{SL}(2,\mathbb{C})$ factorisation.

Because $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$, a projective measurement has at most two orthogonal outcomes. A complementary pair of rays does exist: the coordinate axes of the dual spinor are orthogonal. Four pairwise orthogonal rays do not. The lattice has infinitely many atoms—one-dimensional subspaces, one for each point of the complex projective line—so the Hilbert layer is not a four-valued logic at all. The four D4L labels cannot even enumerate those atoms.

6.4 Quantum logic of $\mathbb{C}^2$

A quantum proposition is a linear subspace of $\mathbb{C}^2$. Meet is intersection, join is the sum, negation is the orthogonal complement. Finite dimension makes every subspace closed. The resulting lattice is orthomodular: if $A \leq B$ then $A \vee (A^{\perp} \wedge B) = B$. It is not distributive. With $e_0 = (1,0)$, $e_1 = (0,1)$, $d = (1,1)$, and $\ell_v = \mathbb{C}v$,

$$\ell_{e_0} \wedge (\ell_{e_1} \vee \ell_d) \neq (\ell_{e_0} \wedge \ell_{e_1}) \vee (\ell_{e_0} \wedge \ell_d) :$$

the left side is ℓ_{e_0}, the right side is $\{0\}$.

The scalar connectives of Section 4 distribute. A distributive lattice cannot be isomorphic to the subspace lattice of a Hilbert space of dimension at least two. There do not exist three pairwise orthogonal one-dimensional subspaces of $\mathbb{C}^2$. The four D4L labels therefore cannot be four orthogonal rays, and the amplitude connectives cannot present the Birkhoff–von Neumann lattice.

6.5 Projectors and the Dirac spinor

A primitive idempotent in the algebra must be built from generators that square to $+1$. If $g^2 = +1$, then $(1+g)/2$ is idempotent. Write $j := e_0 e_1$ for the hyperbolic generator of axis 0 (so $j^2 = +1$). This j is not the time vector $\gamma^0 = e_0$, which squares to -1 in signature $(-,+,+,+)$. The working projectors

$$P_{\text{spin}} = \frac{1 + e_0 e_1}{2}, \qquad P_L = \frac{1 - e_1}{2}, \qquad P_R = \frac{1 + e_1}{2}$$

are each idempotent, and $P_L + P_R = 1$. The formulae $(1 \pm i)/2$ built from the $\mathrm{Cl}(3,1)$ pseudoscalar i with $i^2 = -1$ are not: $\big((1-i)/2\big)^2 = -i/2$. The product $P_{\text{spin}} P_R$ is likewise not idempotent, because $e_0 e_1$ and e_1 anticommute.

A commuting square-$+1$ pair does exist: e_1 and $e_0 e_2$. Their product projector

$$P = \frac{1 + e_1}{2} \cdot \frac{1 + e_0 e_2}{2}$$

is idempotent and nonzero. The principal left ideal it generates has real dimension exactly eight. It is the direct sum of the four-dimensional left ideal $\mathrm{Cl}(3,1)\,P$ in the Minkowski subalgebra and a copy of that ideal along the null generator e_4; the copy is a proper submodule for $G(3,1,1)$, so the eight-dimensional ideal is reducible as a PGA-module. The number eight matches the real dimension of a Dirac spinor $\mathbb{C}^4$, but the match is doubling along the null direction, not evidence that the eight-dimensional ideal is an irreducible Dirac module. Irreducibility of the Minkowski core as a $\mathrm{Cl}(3,1)$-module, and identification of either ideal with the matrix representation, are not claimed.

The Dirac matrices of signature $(-,+,+,+)$ are the grade-1 generators γ^μ of $\mathrm{Cl}(3,1)$. They obey

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \qquad \eta = \mathrm{diag}(-1,+1,+1,+1).$$

In particular $(\gamma^0)^2 = -1$, so $\gamma^0 \neq j$. The same Clifford relations are realised by matrices in $M_4(\mathbb{C})$ acting on a Dirac spinor space $\mathbb{C}^4$. That space has complex dimension four, hence is not the dual spinor $\mathbb{C}^2$; as real vector spaces one has a Weyl splitting $\mathbb{C}^4 \simeq \mathbb{C}^2 \oplus \mathbb{C}^2$. The algebraic reason for the split is chirality. With $\gamma^5 = -i\gamma^0\gamma^1\gamma^2\gamma^3$ one has $(\gamma^5)^2 = I$ and $\{\gamma^5, \gamma^\mu\} = 0$. The Weyl projectors

$$P_{\pm} = \frac{1 \pm \gamma^5}{2}$$

are complementary idempotents. The upper Weyl block is the $+1$ eigenspace of γ^5; the lower block is the -1 eigenspace. The splitting is a dictionary, not an identification of the dual-sector kinematics with a chiral Dirac component.

The same Clifford product already contains a first-order factor of the Minkowski wave operator. For a four-momentum p write

$$Q(p) = -p_0^2 + p_1^2 + p_2^2 + p_3^2, \qquad \gamma(p) = \sum_\mu p_\mu \gamma^\mu.$$

The relations of $\mathrm{Cl}(3,1)$ give $\gamma(p)^2 = Q(p)$ in the algebra, and the same identity as matrices on $\mathbb{C}^4$. The mass shell $Q(p) = -m^2$ is the vanishing of a quadratic that factors as $(\gamma(p) - im)(\gamma(p) + im)$. A spinor with $\gamma(p)\Psi = im\Psi$ therefore lies on that shell: this is the Dirac equation in momentum space, not an extra postulate. In the Weyl representation $\gamma(p)$ is off-diagonal. Vanishing mass decouples the two chiral blocks; a nonzero mass mixes them. The two copies of $\mathbb{C}^2$ that make up a Dirac spinor are the two chiral halves of one massive fermion, not two independent particles. Chirality γ^5 sends a solution of mass m to a solution of mass $-m$. Rest-frame solutions exist. A spacetime PDE, covariance under the full rotor group, and the identification of m with torsional lag are not claimed.

6.6 Internal dictionary

The two layers share a geometry. They do not share operations.

Amplitude layer	Dual Hilbert layer
pure dual amplitude	$\beta \in \mathbb{R}^3$ and $U(\beta) \in \mathrm{SU}(2)$
pure usual amplitude	$\alpha \in \mathbb{R}^3$ and Hermitian det $= 1$ boost $V(\alpha)$
usual–dual swap	leaves the dual sector; not a Hilbert adjoint
Killing overlap	distinct from the spinor inner product
vanishing interference on one axis	commuting generators (necessary only)
J_{norm} four-state collapse	coarse-graining of a continuous scalar
min / max	not the meet/join of $L(\mathbb{C}^2)$
dual $\mathbb{C}^2$	not the Dirac $\mathbb{C}^4$

There is no assignment of one-dimensional subspaces of $\mathbb{C}^2$ to the four D4L labels that makes distinct labels pairwise orthogonal. The four names are a partition of an interval, not a projective measurement of a qubit.

7 Which heights hold

The four labels are not yet a logic until one says which heights *hold*. D4L uses two designated predicates on $J_{\mathrm{norm}} \in [-1, 1]$, not one.

Definition 7.1 (Designated values)**.**

$$\mathrm{HoldsT}(j) \iff j = 0, \qquad \mathrm{HoldsNotF}(j) \iff j < 1.$$

The first is synchrony (the label $|T\rangle$). The second is non-refutation (every label except $|F\rangle$).

The two predicates answer different questions. Synchrony asks whether the usual and dual frames are balanced. Non-refutation asks only whether the configuration has been driven onto the unique all-boost world. A rotation-dominant configuration is not synchronous, but it is not refuted either.

The designated point is the information bottom, not the top of the height order, and the two predicates read the connectives in opposite directions. Non-refutation of a conjunction is the disjunction of the non-refutations: a meet reaches the positive wall only when both heights already sit there. Non-refutation of a disjunction is the conjunction of the non-refutations: a join reaches that wall as soon as either argument does. Synchrony is stricter, and it absorbs. The conjunction of balance with any nonnegative height, the positive wall included, is balance again; the disjunction of balance with any nonpositive height is balance again. In particular $0 \wedge 1 = 0$: the positive wall does not absorb conjunction.

Formulas are generated by atoms, $\neg$, $\wedge$, and $\vee$, and are evaluated by the scalar operations already fixed on heights. An implication on those heights would have to answer both designated predicates at once. Each predicate has a residual of its own. The synchrony residual returns the consequent when the antecedent is balanced, and balance otherwise. The non-refutation residual returns balance when

the antecedent sits on the positive wall, returns that wall when a non-refuted antecedent implies it, and otherwise returns the consequent.

$$j \to_T k = \begin{cases} k & \text{if } j = 0, \\ 0 & \text{otherwise,} \end{cases} \qquad j \to_{\neg F} k = \begin{cases} 0 & \text{if } j = 1, \\ 1 & \text{if } j < 1 \text{ and } k = 1, \\ k & \text{otherwise.} \end{cases} \tag{3}$$

Thus $j \to_T k$ is synchronous exactly when a synchronous antecedent forces a synchronous consequent, and $j \to_{\neg F} k$ is non-refuted exactly when a non-refuted antecedent forces a non-refuted consequent.

Theorem 7.2 (No common residual)**.** *No binary operation on $[-1, 1]$ is synchronous exactly when a synchronous antecedent forces a synchronous consequent, and non-refuted exactly when a non-refuted antecedent forces a non-refuted consequent.*

Any operation that meets the synchrony clause sits at balance as soon as the antecedent is off balance, because the conditional is then vacuous. Any operation that meets the non-refutation clause sits on the positive wall as soon as the antecedent is off that wall and the consequent is on it. The clauses meet when a height other than balance or the wall implies the wall. A boost-dominant witness is $\frac{1}{2} \to_T 1 = 0$ and $\frac{1}{2} \to_{\neg F} 1 = 1$. Synchrony lies off the wall, so the two readings cannot share a value.

The Gödel residuum of min, with lattice top 1, preserves the top of the height order and reads that top as truth. A boost-dominant height implying synchrony is sent to balance, in agreement with both residuals. A rotation-dominant height implying synchrony lies below the consequent, so the residuum returns the wall. The non-refutation residual returns balance there.

Both residuals send a boost-dominant height implying synchrony to balance, and every synchrony residual must. Non-refutation leaves that successful value free, forbidding only the wall. The status algebra uses the freedom (Section 11). The two residuals return the same height precisely when the antecedent is balanced, or sits on the positive wall, or the consequent is balanced.

A valuation assigns each atom a height in $[-1, 1]$. Consequence follows the two predicates: $\Gamma \models_T \psi$ when every valuation that synchronises Γ synchronises ψ; $\Gamma \models_{\neg F} \psi$ is the same clause with non-refutation. The two relations are not the same. $P \wedge \neg P$ is non-refuted under every valuation, so the empty set non-refutes it; it is synchronous only when P is, so the empty set does not T-entail it. Synchrony of an atom does force synchrony of $P \vee \neg P$. Non-refutation does not: the negative wall is not refuted, and the disjunction of that wall with its swap is the positive wall.

Two-valued *restrictions* of the interval are not unique, because D4L's names $|T\rangle, |F\rangle$ are not classical true/false. The *named* fragment is $\{0, 1\} = \{|T\rangle, |F\rangle\}$. The *wall* fragment is $\{-1, +1\} = \{\text{deepest } |B\rangle, |F\rangle\}$. They behave differently under negation. On the named fragment, $\neg 0 = 0$: the balanced point is a fixed point of negation, and $\neg 1 = -1$ leaves $\{0, 1\}$ altogether. The named fragment is therefore not Boolean and not even closed. On the walls, $\neg$ swaps the two points and has no fixed point. The wall fragment is the faithful classical core, in the sense fixed in Section 8. Comparing D4L with "true and false" by cutting to the names $|T\rangle$ and $|F\rangle$ already leaves classical logic.

8 Classical proof on the walls

Read the positive wall as classical truth and the negative wall as classical falsity. Negation swaps the two points. Conjunction and disjunction are min and max, and both preserve the walls, so a formula whose atoms sit on the walls evaluates on the walls. Under that reading the wall fragment is a Boolean algebra.

Theorem 8.1 (Classical tautologies)**.** *A formula built from atoms, negation, conjunction, and disjunction is a classical tautology if and only if every assignment of its atoms to the two walls returns the positive wall.*

A classical derivation spends most of its length on identities that do not use this restriction. Section 4 already records that De Morgan duality and distributivity hold for every real height. Double negation and absorption are of the same kind. A proof of any of them by cases on true and false repeats a computation the height has already performed. The case split records the specialisation of that identity.

The walls contribute saturation. Section 4 records $j \wedge \neg j = -|j|$ and $j \vee \neg j = |j|$. Both equal the corresponding wall if and only if the height is already a wall. Off the walls the comparison $-|j| \leq |k|$ remains: the conjunction of any height with its negation lies at or below the disjunction of any height with its negation. Excluded middle, and the classical form of non-contradiction, are the statement $|j| = 1$.

Material implication, read as $\neg j \vee k$, shows the split on a syllogism. The formula

$$(P \to Q) \to \big((Q \to R) \to (P \to R)\big)$$

returns the positive wall under every wall assignment. It is not an identity of the height. At total balance, every atom at 0, the same formula returns 0. Synchrony holds. The classical reading does not call that value truth. One computation, two readings, and they disagree on whether the inference has succeeded. A classical proof of the syllogism is a wall computation. The step that stays on the walls is the saturation $|j| = 1$. The De Morgan skeleton underneath it has already been proved on the height.

The inference that uses the implication is detachment. Write $j \to k$ for $\max(-j, k)$.

Theorem 8.2 (Three cuts)**.** *For all real heights, $j = 1$ and $j \to k = 1$ imply $k = 1$. The implication lies strictly below the positive wall if and only if $-j$ and k both do. If $k \leq 0$, then $0 \to k = 0$.*

The first cut is classical detachment, and it holds for every real height. Whenever the antecedent and the implication both sit at the positive wall, the consequent sits there with them. The second cut is non-refutation of the implication. It is non-refutation of the negated antecedent together with non-refutation of the consequent. Non-refutation of the antecedent itself is the different inequality $j < 1$. The two inequalities agree on the open interval $(-1, 1)$ and separate on the walls. The witness is the negative wall implying balance: $-1 \to 0 = 1$. Both heights lie strictly below the positive wall, and the implication is that wall. Classical success and the failure of non-refutation are one value. The third cut is synchrony. Balance materially implies every nonpositive height, and the implication is itself balance. Whenever the consequent is strictly negative, both premises of detachment are synchronous and the consequent is not. A classical proof is written in the implication that keeps all three cuts. The residuals of Section 7 restore one cut at a time, and they agree only on the locus already named there.

Explosion has the same shape. On the walls a value and its negation cannot both be the positive wall, so the inference from a contradiction to an unrelated atom quantifies over an empty set of valuations. The classical proof is short because the premise does not occur.

9 What two-valued logic cannot host

Section 8 factors a classical proof into a height identity, a wall saturation, and the three cuts of detachment. Two-valued algebras still cannot host the situations below, and D4L still realises them. The difference is fixed as a pair: emptiness (or vacuity, or collapse to a singleton) on the fragment, realisation on D4L.

These are changes of semantics, not new theorems of Peano arithmetic. Unconditional FLT / Beal and a refutation of Gödel's theorems are not claimed.

9.1 A fixed point of negation

Classically, $P \leftrightarrow \neg P$ has no model. On the wall fragment the same is true: if $j \in \{-1, +1\}$ then $j \neq -j$. In D4L the condition $j = \neg j$ is equivalent to $j = 0$. The unique solution is the label $|T\rangle$, perfect usual–dual balance, and it is realised both by a constant valuation and by an amplitude.

The equation is not the Gödel sentence. It is the self-referential *equality* $P = \neg P$, which an algebra whose negation has no fixed point cannot host, and which D4L seats uniquely at synchrony.

9.2 Contradiction without explosion

Classical two-valued logic has *ex falso quodlibet*: from $\{P, \neg P\}$ every Q follows. On Bool the premise is already impossible, because P and $\neg P$ cannot both be `true`. On the walls, j and $\neg j$ cannot both be $+1$. Two-valued explosion is therefore vacuous: the premises cannot be designated together.

In D4L they can. Let P be atom 0 and Q atom 1. Then

$$\{P, \neg P\} \not\models_T Q, \qquad \{P, \neg P\} \not\models_{\neg F} Q.$$

One witness sends P to 0 and Q to 1: the contradictory pair is synchronous, the unrelated atom is False. A second witness sends P to $-\frac{1}{2}$ (interior $|B\rangle$) and Q to 1: the pair is non-refuted, the unrelated atom is still False. The conjunction $j \wedge \neg j$ is always non-refuted. An inconsistent pair of observations therefore does not force an unrelated atom onto the positive wall.

9.3 The constraint $J_{\text{norm}} < 1$

A sentence forbidden from sitting at the unique all-boost world is constrained by HoldsNotF, that is by $J_{\text{norm}} < 1$. On the named fragment this constraint collapses to HoldsT: the only remaining point is 0. On D4L the same constraint is inhabited by three labels, $|T\rangle$, $|U\rangle$, and $|B\rangle$. Non-refutation is not synchrony. The named fragment cannot see the difference.

9.4 Complementarity

The subspace lattice of $\mathbb{C}^2$ is not distributive, hence not a Boolean algebra. It has at most two pairwise orthogonal atoms. The four D4L labels are not a PVM. The distributive scalar connectives cannot present that lattice. Complementary rays in the dual Hilbert layer cannot be assigned Boolean values, and they cannot be identified with the four names of the amplitude layer. The two layers remain two layers.

10 Where a classical proof survives in the spinor lattice

The subspace lattice of $\mathbb{C}^2$ is not Boolean, so a classical proof about subspaces is not yet a proof in the dual Hilbert layer. The part that survives has a geometric name.

The implication carried by the orthomodular law is

$$A \to B \;=\; A^{\perp} \vee (A \wedge B).$$

The zero subspace implies every subspace, and no nonzero vector lies in the zero subspace. Quantum explosion, like the explosion of Section 8, is an inference from the bottom of the lattice.

Theorem 10.1 (Inclusion, splitting, and detachment)**.** *$A^{\perp} \vee (A \wedge B)$ equals the whole space if and only if A is contained in B. The following are equivalent: A splits along B,*

$$A \;=\; (A \wedge B) \vee (A \wedge B^{\perp});$$

the lattice implication equals the classical expansion $A^{\perp} \vee B$; and the expansion detaches,

$$A \wedge (A^{\perp} \vee B) \;\subseteq\; B.$$

An inclusion is the first identity. Splitting is the further condition that compares the lattice implication with the classical expansion. The lattice implication sits inside that expansion, because $A \wedge B$ sits inside B. Equality is the splitting, and detachment of the expansion is the same condition. The implication the lattice carries detaches for every pair: a vector that lies in A and in $A^{\perp} \vee (A \wedge B)$ already lies in $A \wedge B$, hence in B. On a splitting pair the expansion is this implication, so the same vector lies in B. The relation is symmetric: A splits along B exactly when B splits along A. A pair that splits lies in a Boolean block, and inside that block the classical expansion is the implication the lattice already uses.

Orthogonal rays of a computational basis are such a pair. Each is the orthogonal of the other, the two implications agree, and both equal the orthogonal line. The agreed value is that line. A ray sits outside the orthogonal ray, so the implication stops short of the whole space.

A computational ray and the diagonal do not split. Their meet is zero, and the ray is not orthogonal to the diagonal, so the decomposition is the zero subspace. The classical expansion returns the whole space. The implication carried by the lattice returns only the orthogonal line. The meet of the ray with the expansion is the ray, and the ray does not lie in the diagonal, so the expansion does not detach. That pair is the step at which a two-valued proof leaves the lattice.

11 Proof status and Diophantine regimes

A DST proof artefact—a core lemma, a live bridge, a refuted diagnostic, a bookkeeping identity—is not a torsion configuration. It carries a discrete status that reuses the four names, but the observable is not J_{norm}. On this carrier, negation *is* a function of the four labels, and implication is a table, not the residuum of min.

Negation and the lattice operations follow the amplitude representatives $|T\rangle \mapsto 0$, $|U\rangle \mapsto 1/2$, $|F\rangle \mapsto 1$, $|B\rangle \mapsto -1/2$. In particular $F \wedge \neg F = B$. A second operation, the establishedness meet, packages two

artefacts along $F < B < U < T$, so that a proved core with a live bridge is still live: the meet of T and U is U. Implication is

$$T \to \phi = \phi,$$
$$F \to \phi = T,$$
$$U \to T = U,$$
$$B \to T = U.$$

The last two clauses are the point. The implication is refuted exactly when a non-refuted status implies a refuted one, so the table residuates non-refutation. An open or conflicted status implying a theorem stays off T. Every synchrony residual on the height would have placed the open clause at balance. No Belnap-FOUR isomorphism is claimed. Designated predicates on statuses reuse the same two names: HoldsT_R means status T; $\mathrm{HoldsNotF}_R$ means status $\neq F$.

This is the algebra in which DST records what it has proved and what it has not. Named two-valued logic $\{T, F\}$ cannot host a live status U. The wall fragment $\{B, F\}$ cannot host a proved core T. D4L realises the assignment core T, diagnostic F, live U, conjecture U. The core alone does not T-entail the conjecture. A window T together with an ill-posed NoGo B does not force the conjecture off F. The implication $U \to T$ is not designated.

The same algebra hosts a Beal residual atlas: closed slices (including the Fermat and Darmon–Merel positions, and the signature $(n, n, 5)$) sit at T; refuted diagnostics sit at F; bookkeeping identities sit at B; live residuals and the classical conjecture sit at U. The closed slices do not T-entail classical Beal. Even permutations of the cube and fifth-power signatures stay in the even two-equal difference residual when the common exponent is not divisible by three: an even exponent supplies no sign that would carry them onto Darmon–Merel or $(n, n, 5)$ form. When that common exponent is divisible by three, the equation returns to a coprime sum of two cubes and the signature closes for every remaining exponent at which such a sum is never a perfect power. Mihăilescu's consecutive-power theorem closes a coefficient condition without closing the exponent shape that hosts it. The status board does not encode the arithmetic assembly of the residuals. Forcing every named artefact to T still fails to T-entail the conjecture; promoting a single live residual from U to T likewise cannot claim it. The exponent-shape classifier lands in $\{T, U, F\}$ and never records bookkeeping B. A dual-axis Fermat atlas likewise places the additive core, the real L^p dichotomy, and the elliptic finite sandwich of N_1 at T, demoted single-axis modular and balanced-seed diagnostics at F, and the mixed-motor residual together with classical FLT at U. Closed Fermat slices do not T-entail classical FLT.

This is why DST must not treat a single bound, or a refuted diagnostic, as entailing the seven classical conjectures. The regime layer records that prohibition. It does not prove any unproved bridge, and it does not claim unconditional FLT, Beal, or abc.

Additive motors and multiplicative amplitudes occupy distinct seats at the same label $|T\rangle$. Pure translation has vanishing torsion and is vacuum. A balanced massive configuration is not vacuum. The mixed product $R \cdot T$ is not identified with $\exp(\Omega_{\mathrm{biv}})$ when the factors fail to commute. The balanced residual class excludes window seeds: a half-window configuration is not a balanced residual.

On the finite torus the same four names acquire a discrete reading. When $4 \nmid N$ the sharp discrete bound is strict, so no admissible point saturates $|F\rangle$. When $4 \mid N$ both walls are attained: the all-boost lattice point realises $|F\rangle$, and the all-rotation lattice point realises deepest $|B\rangle$. Pure-boost scaling multiplies J_{norm} by k^2. Synchrony is a fixed point. A label $|U\rangle$ can be driven onto $|F\rangle$, or out of the admissible cone, where collapse is undefined. That is the logical reading of the shared no-go; it is not new arithmetic.

Vacuum is a fixed point of every admissible scaling. A small balanced-massive seed stays at $|T\rangle$ with mass $k^2 M$ and may remain admissible; a large balanced seed leaves the cone. That cone exit is invisible to the signed-height classifier: J_{norm} stays 0, so the four-state label is still $|T\rangle$. Mass and the cone predicate are the observables that see it. This is why a Beal balanced seed $\theta = \log 2/m$ can survive m-fold scaling inside the cone.

Height and modular winding are two measurements. Real-scale admissibility and nonzero winding cannot hold together. That incompatibility is a separation of observables, not a Both-status on one observable.

12 A reading of self-reference

Let G be the Gödel sentence: "G is not provable in the system." In D4L, provability is read as a drive toward synchrony $J_{\mathrm{norm}} \to 0$, i.e. toward $|T\rangle$. The honest constraint on a sentence forbidden from sitting

at the unique all-boost world is exactly HoldsNotF of Section 9.3:

$$J_{\text{norm}}(G) < 1.$$

That constraint leaves G in $|T\rangle$, $|U\rangle$, or $|B\rangle$ as labels, or at $|T\rangle$ if one further demands synchrony. Conjunction does not explode (Section 9.2).

This is a change of semantics, not a proof that Gödel's theorems fail in classical arithmetic. Completeness talk over the negative-torsion sector is likewise a reading of admissible rotors, not a completion of the Hilbert programme inside first-order Peano arithmetic. The content of Section 9 is fixed; the reading in this section is not a new incompleteness theorem.

13 Scope

D4L takes admissible torsion configurations as the carriers of propositions and J_{norm} as the observable. It partitions $[-1, 1]$ into four named states, both walls attained on the admissible cone, and it makes negation the usual–dual swap of the geometry. Conjunction of a height with its negation stays out of $|F\rangle$. Conjunction with balance returns to balance, and non-refutation reverses meet and join. Height and saturation are two preorders on one interval. Height cannot exceed mass, with equality only when one sector vanishes. The Killing pairing is indefinite; self-overlap is not a Born probability; the dual sector yields a Euclidean three-space, a quaternion table, and an SU(2) rotor on $\mathbb{C}^2$ whose closed-subspace lattice is orthomodular and non-distributive, with infinitely many atoms and a complementary pair of rays but not four. The usual sector is the twin: hyperbolic Rodrigues, a Hermitian determinant-1 matrix rotor that is not unitary for a nonzero boost. Chirality γ^5 squares to the identity, anticommutes with the Dirac matrices, and splits $\mathbb{C}^4$ by complementary Weyl projectors. The Minkowski quadratic is a square in $\mathrm{Cl}(3, 1)$, so a momentum-space spinor with $\gamma(p)\Psi = im\Psi$ lies on the mass shell $Q(p) = -m^2$; mass mixes the Weyl blocks, and γ^5 reverses the sign of m.

Two designated predicates and two two-valued fragments turn that picture into a consequence relation. The relation hosts a unique negation fixed point at $|T\rangle$, refuses explosion of $\{P, \neg P\}$, inhabits $J_{\text{norm}} < 1$ by three labels, and records the DST three-layer split—proved core, refuted diagnostic, open bridge—without T-entailing a classical conjecture. No operation on the height residuates both designated predicates. Implication lives on the status algebra, where the table residuates non-refutation and $U \to T = U$. Classical tautologies are the formulas that return the positive wall on every wall assignment. The De Morgan skeleton holds for every real height; excluded middle and the syllogism are the saturation $|j| = 1$, and at balance the syllogism returns synchrony. Material implication detaches the positive wall on every real height, lies below that wall exactly when the negated antecedent and the consequent do, and returns balance when balance implies a nonpositive height. In the spinor lattice, $A^\perp \vee (A \wedge B)$ is the whole space exactly when A is contained in B. It agrees with $A^\perp \vee B$, and $A \wedge (A^\perp \vee B)$ lies in B, exactly when A splits along B. The lattice implication detaches for every pair. A basis ray and the diagonal separate the two implications, and on that pair the expansion does not detach. Label $|T\rangle$ is not a single geometric point: vacuum, axiswise balanced massive, and cross-axis cancellation all sit there, distinguished by mass and by axiswise equality, without a fifth name. Discrete $|F\rangle$ is forbidden when $4 \nmid N$; when $4 \mid N$ both walls are attained. Amplification is a partial map on labels, including a cone exit of balanced massive that the height classifier cannot see.

The theory does not identify amplitude operations with Hilbert operations, four labels with a PVM, vanishing interference with a common single axis, or raw J with a value in $[-1, 1]$. It does not derive the Born rule, a spacetime Dirac PDE, or a solution of the measurement problem. The momentum-space Dirac equation is the first-order factor of the Minkowski quadratic recorded in Section 6.5; it is not a derivation of mass from torsion. It records that the commuting left ideal is reducible as a $G(3, 1, 1)$-module by null doubling. It does not claim $\mathrm{Cl}(3, 1)$ irreducibility of the Minkowski core, nor identify either ideal with the matrix Dirac module, nor identify the Banach exponential with the matrix exponential beyond a single pure sector, nor factor mixed usual/dual rotors through $\mathrm{SL}(2, \mathbb{C})$. It does not refute Gödel's theorems in classical arithmetic, and it does not claim unconditional Beal, FLT, or abc.

The statements above are machine-checked.

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